Secure Campus Network

Using Cisco Packet Tracer • Mini Project

Router Configuration • Switch Security • SSH • Port Security • Console/Enable Security • Password Encryption

Practical step-by-step documentation

1. Project Objective

Design and implement a small secure campus network in Cisco Packet Tracer. The project covers basic Cisco device configuration and security mechanisms: router and switch configuration, console and privileged EXEC security, SSH remote access, switch port security, and password encryption. Each step below includes the commands used and a screenshot taken from the working Packet Tracer file (MiniProject.pkt).

2. Network Scenario

A small university department has one router, two Cisco switches and six PCs. The router connects two IP networks (192.168.10.0/24 and 192.168.20.0/24). The switches provide LAN connectivity and are secured against unauthorized access and unauthorized devices.

> **Note on device names:** A Cisco 1941 router was used in Packet Tracer (labelled Router0, configured with hostname R1). The switches are 2960-24TT models (Switch0 = S1, Switch1 = S2), and the PCs are labelled PC0–PC5, which correspond to PC1–PC6 in the addressing plan.

3. IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.10.1	255.255.255.0	—
R1	G0/1	192.168.20.1	255.255.255.0	—
S1	VLAN 1	192.168.10.2	255.255.255.0	192.168.10.1
S2	VLAN 1	192.168.20.2	255.255.255.0	192.168.20.1
PC1 (PC0)	NIC	192.168.10.10	255.255.255.0	192.168.10.1
PC2 (PC1)	NIC	192.168.10.11	255.255.255.0	192.168.10.1
PC3 (PC2)	NIC	192.168.10.12	255.255.255.0	192.168.10.1
PC4 (PC3)	NIC	192.168.20.10	255.255.255.0	192.168.20.1
PC5 (PC4)	NIC	192.168.20.11	255.255.255.0	192.168.20.1
PC6 (PC5)	NIC	192.168.20.12	255.255.255.0	192.168.20.1

4. Physical Connections

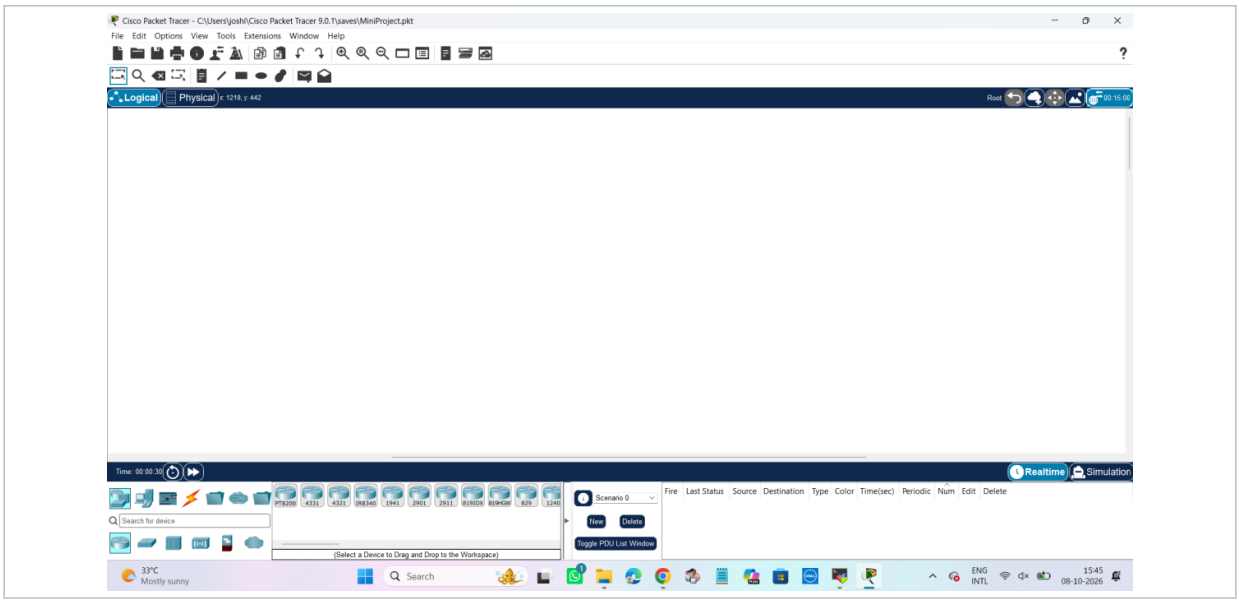
- R1 G0/0 → S1 G0/1
- R1 G0/1 → S2 G0/1
- S1 Fa0/1 → PC1, Fa0/2 → PC2, Fa0/3 → PC3
- S2 Fa0/1 → PC4, Fa0/2 → PC5, Fa0/3 → PC6
- All links use copper straight-through cables.

Part 1 — Build the Topology

Place the devices in Packet Tracer and connect them.

Step 1 — Add the router

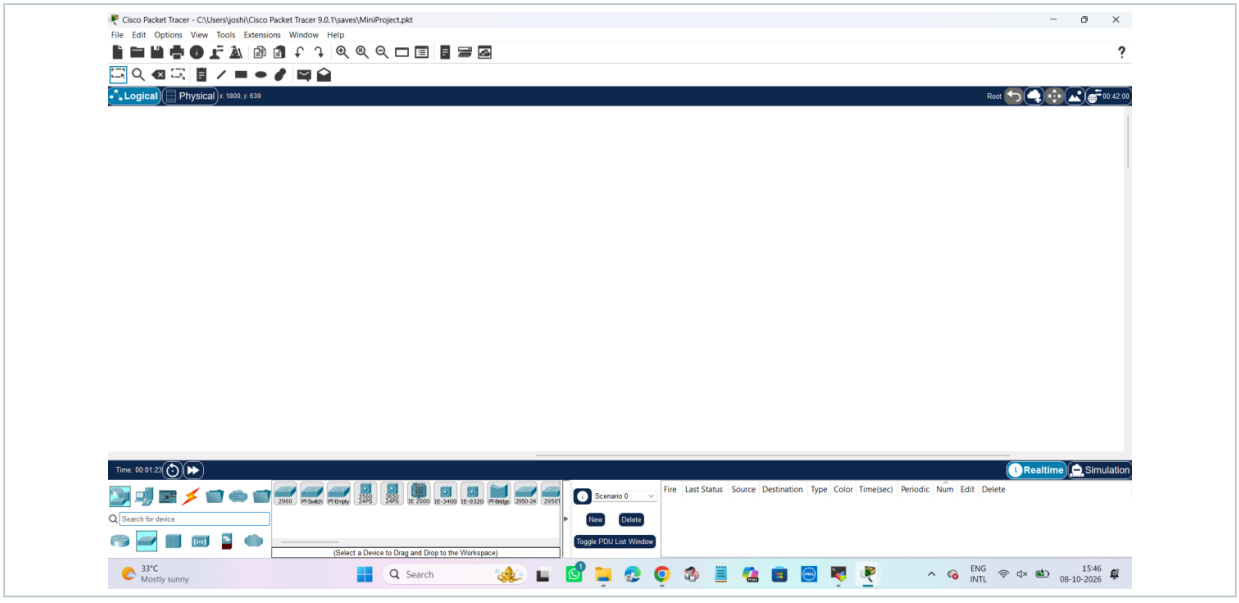
Open a new Packet Tracer workspace and save it as **MiniProject.pkt**. In the device palette choose **Routers** and drag a router onto the workspace. The **1941** model was used for this project.



What this step demonstrates: Start the topology by placing the router (R1) from the Routers category.

Step 2 — Add the two switches

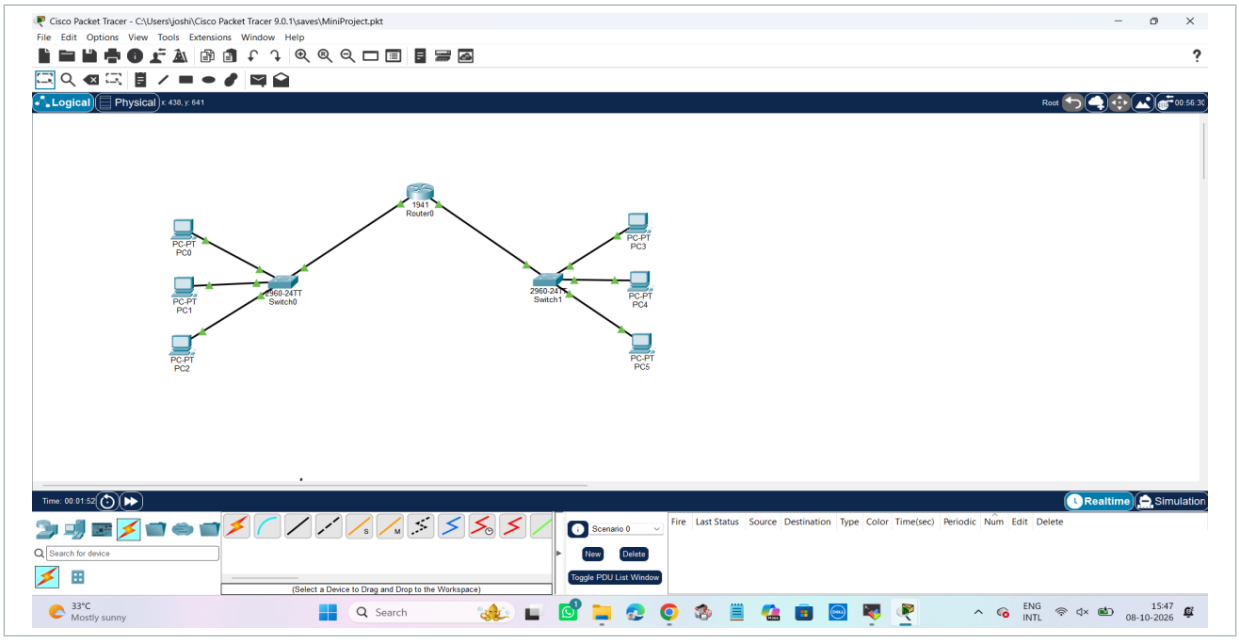
Select the **Switches** category and drag two **2960** switches onto the workspace, one on each side of the router. These become **S1** (Switch0) and **S2** (Switch1).



What this step demonstrates: Place the two access-layer switches that will connect the PCs.

Step 3 — Add the PCs and connect all devices

Add six PCs from **End Devices** (three per switch). Using the **Copper Straight-Through** cable, connect the devices exactly as listed in the physical-connections plan: router G0/0 to S1 G0/1, router G0/1 to S2 G0/1, and each PC to Fa0/1, Fa0/2 and Fa0/3 of its switch.

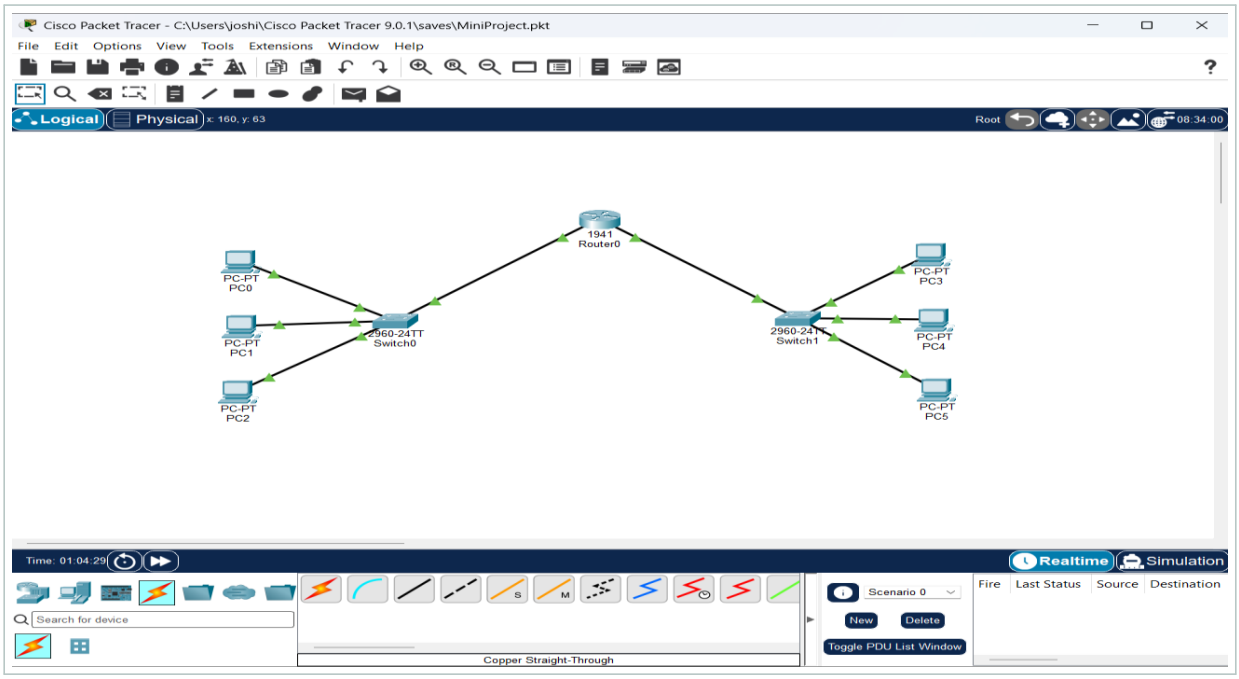


What this step demonstrates: Complete the physical topology: one router, two switches and six PCs forming two LANs.

Tip: Green link lights may take a few seconds to appear while the ports come up.

Step 4 — Verify the topology

The finished topology in the saved **MiniProject.pkt** file shows **Router0** in the centre, **Switch0** with PC0–PC2 on the left (Network 1) and **Switch1** with PC3–PC5 on the right (Network 2). All link indicators are green.



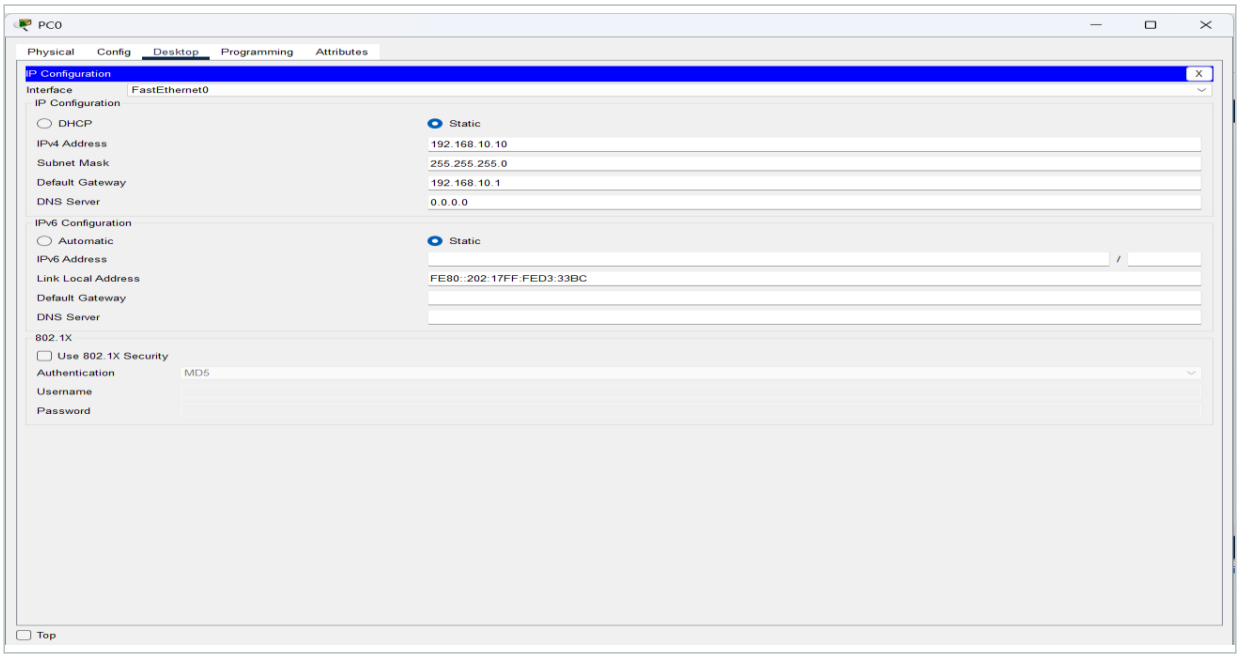
> **What this step demonstrates:** Confirm that all cables are connected and every link is up before configuring the devices.

Part 2 — Configure the PCs

Assign static IP addresses using Desktop → IP Configuration.

Step 5 — Configure PC0 (PC1 in the plan)

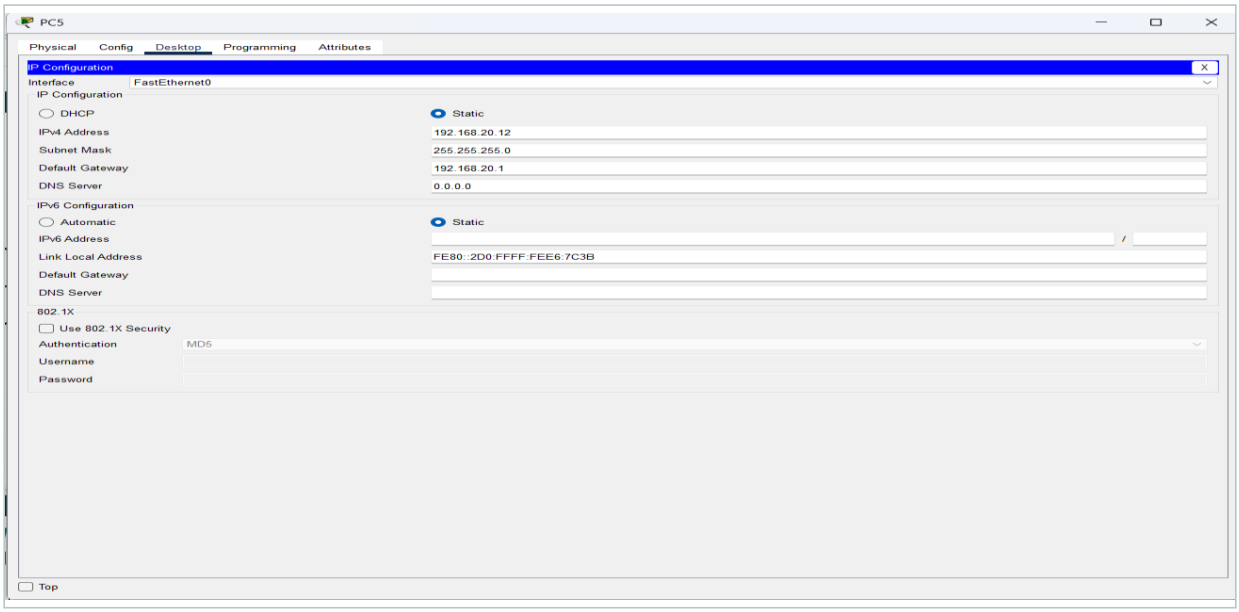
Click the PC, open the **Desktop** tab and select **IP Configuration**. Choose **Static** and enter IPv4 address **192.168.10.10**, subnet mask **255.255.255.0** and default gateway **192.168.10.1**. PC0 is connected to S1 Fa0/1.



What this step demonstrates: Give the first PC on Network 1 its static address and the router G0/0 address as default gateway.

Step 6 — Configure PC5 (PC6 in the plan)

Repeat the same procedure for every PC. PC5 on Switch1 uses IPv4 address **192.168.20.12**, mask **255.255.255.0** and default gateway **192.168.20.1** (router G0/1). The other PCs follow the addressing table: PC1 **192.168.10.11**, PC2 **192.168.10.12**, PC3 **192.168.20.10** and PC4 **192.168.20.11**.



> **What this step demonstrates:** Give the PCs on Network 2 their static addresses with the router G0/1 address as default gateway.

Part 3 — Configure and Secure Router R1

Interfaces, console and enable security, password encryption and SSH.

Step 7 — Configure the router interfaces (Task 1)

Open **Router0** → **CLI** and configure the hostname and both Gigabit interfaces, then check them with **`show ip interface brief`**.

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
show ip interface brief
```

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

In this capture, **GigabitEthernet0/0** is configured with **192.168.10.1** and shows **up/up**. GigabitEthernet0/1 is configured with 192.168.20.1 in the same way; the successful ping to Network 2 in Step 17 confirms it is working.

> **What this step demonstrates:** Bring up the two router interfaces that act as the default gateways for the two networks.

Step 8 — Secure the console and enable mode, and encrypt passwords (Tasks 2 and 3)

Protect console access with a line password and privileged EXEC mode (**R1#**) with an **enable secret**. Then turn on **service password-encryption** and check the result with **`show running-config`**.

The first attempt to enter **`line console 0`** returned **% Invalid input detected** because it was typed in privileged mode; entering **`enable`** and **`configure terminal`** first fixed it. Likewise, **`show running-config`** must be run from privileged mode (after **`exit`**), which is why it succeeds at the end.

Console and privileged EXEC security

```
line console 0
password Console@123
login
exit
enable secret Enable@123
```

Password encryption

```
service password-encryption
show running-config
```

```
R1#line console 0
          ^
% Invalid input detected at '^' marker.

R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line console 0
R1(config-line)#password Josh!th@957
R1(config-line)#login
R1(config-line)#exit
R1(config)#enable secret Josh!
R1(config)#service password-encryption
R1(config)#show running-config
                ^
% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config
Building configuration...

Current configuration : 712 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$PaNX.wLABqlcK9hPWFsRz.
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
```

The running configuration now contains **service password-encryption** and an **enable secret 5 …** line, where the secret is stored as a hash instead of plain text.

> **What this step demonstrates:** The console password protects console access, and the enable secret protects privileged EXEC mode. **`enable secret`** is preferred over a plain enable password.

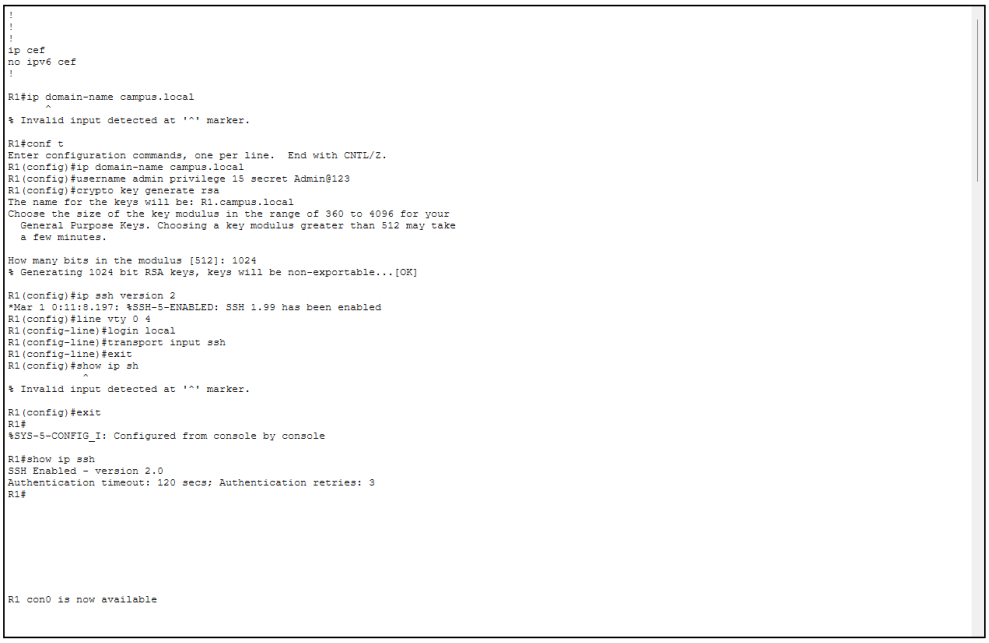
> **Important:** **`service password-encryption`** only obfuscates line passwords in the configuration. It is not strong cryptographic protection, so **`enable secret`** should still be used for privileged access.

Step 9 — Configure SSH on the router (Task 4)

Set a domain name, create a local admin account, generate the RSA keys (enter **1024** when asked for the modulus), enable SSH version 2 and restrict the virtual terminal lines to SSH with local login.

As before, the first **`ip domain-name`** attempt failed in privileged mode and worked after **`conf t`**. After leaving configuration mode, **`show ip ssh`** reports **SSH Enabled – version 2.0**.

```
ip domain-name campus.local
username admin privilege 15 secret Admin@123
crypto key generate rsa
! Enter 1024 when prompted (or 2048 if supported)
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit
show ip ssh
```



What this step demonstrates: Enable encrypted remote administration of the router using SSH version 2 and a local user account.

Part 4 — Configure and Secure the Switches

Management address, console, SSH, port security and unused ports.

Step 10 — Configure Switch S1 (Tasks 5, 6 and 7)

Open **Switch0** → **CLI**. Configure the hostname, enable secret, password encryption and the VLAN 1 management address with its default gateway; secure the console; configure SSH; and enable **sticky port security** on Fa0/1–Fa0/3.

Basic S1 configuration

```
enable
configure terminal
hostname S1
enable secret Enable@123
service password-encryption
interface vlan 1
ip address 192.168.10.2 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.10.1
```

S1 console and SSH

```
line console 0
password SwitchConsole@123
login
exit
ip domain-name campus.local
username admin privilege 15 secret Admin@123
crypto key generate rsa
! Enter 1024 when prompted (or 2048 if supported)
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit
```

Port security on S1

```
interface fastEthernet 0/1
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
interface fastEthernet 0/2
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
interface fastEthernet 0/3
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
```

What this step demonstrates: Secure management access to S1 and limit each PC port to a single learned (sticky) MAC address, shutting the port down on a violation.

Step 11 — Configure Switch S2 (Tasks 8 and 9)

Open **Switch1** → **CLI** and apply the same settings with the Network 2 management address. Port security is applied to Fa0/1–Fa0/3 using an interface range.

Basic S2 configuration, console and SSH

```
enable
configure terminal
hostname S2
enable secret Enable@123
service password-encryption
interface vlan 1
ip address 192.168.20.2 255.255.255.0
no shutdown
exit
ip default-gateway 192.168.20.1
line console 0
password SwitchConsole@123
login
exit
ip domain-name campus.local
username admin privilege 15 secret Admin@123
crypto key generate rsa
! Enter 1024 when prompted (or 2048 if supported)
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit
```

Port security on S2

```
interface range fastEthernet 0/1-3
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
```

> **What this step demonstrates:** Apply the same security baseline to the second switch for Network 2.

Step 12 — Disable unused ports and save the configuration on S1 (Tasks 15 and 16)

Shut down all unused access ports so that no unauthorized device can use them, then save the configuration. The screenshot shows ports Fa0/21–Fa0/24 changing to **administratively down**, followed by **`copy running-config startup-config`** completing with **[OK]**.

```
interface range fastEthernet 0/4-24
shutdown
exit

! Run on R1, S1 and S2
copy running-config startup-config
```

```
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
S1(config-if-range)#exit
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

What this step demonstrates: Reduce the risk of unauthorized devices connecting to unused active ports, and make the configuration permanent.

Step 13 — Verify port security on S1 (Task 13)

Check the MAC address table and the port-security status for Fa0/1 and for all secured addresses:

```
show port-security
show port-security address
show port-security interface fastEthernet 0/1
show mac address-table
```

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.6448.3662    STATIC      Fa0/3
   1    0001.c935.0701    DYNAMIC     Gig0/1
   1    0002.17d3.33bc    STATIC      Fa0/1
   1    00e0.f9d0.bb03    STATIC      Fa0/2
S1#show port-security interface FastEthernet 0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0002.17D3.33BC:1
Security Violation Count   : 0

S1#show port-security address
              Secure Mac Address Table
------------------------------------------------------------------------
Vlan    Mac Address       Type                       Ports   Remaining Age
                                                             (mins)
----    -----------       ----                       -----   -------------
   1    0002.17D3.33BC    SecureSticky               Fa0/1        -
   1    00E0.F9D0.BB03    SecureSticky               Fa0/2        -
   1    0001.6448.3662    SecureSticky               Fa0/3        -
------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 1024
S1#
```

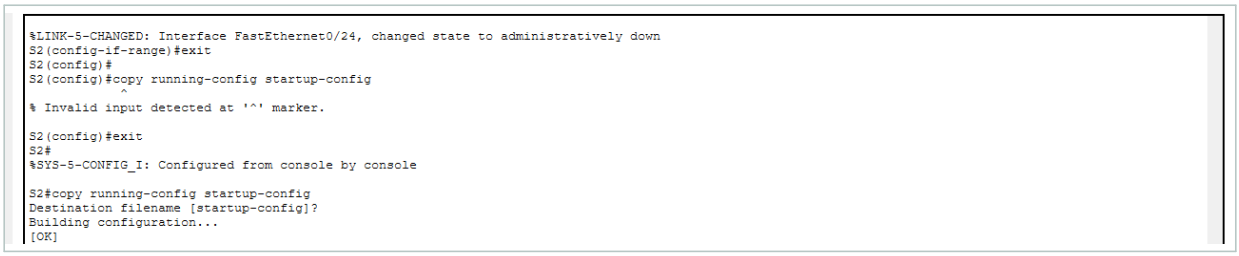
Observations: On Fa0/1 port security is **Enabled**, the port status is **Secure-up**, the violation mode is **Shutdown**, the maximum number of MAC addresses is **1**, and one sticky MAC address (0002.17D3.33BC, PC0) has been learned. The violation count is **0**. The secure address table lists one **SecureSticky** address on each of Fa0/1, Fa0/2 and Fa0/3, and the router is seen dynamically on Gig0/1.

What this step demonstrates: Confirm that port security is enabled, the maximum is 1, sticky learning is active and the violation mode is shutdown.

Step 14 — Disable unused ports and save the configuration on S2

The same two tasks were repeated on S2. Here the first **`copy running-config startup-config`** was typed in configuration mode and returned **% Invalid input**; after **`exit`** to privileged mode the command ran successfully and saved the configuration.

```
interface range fastEthernet 0/4-24
shutdown
exit
exit
copy running-config startup-config
```



```
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
S2(config-if-range)#exit
S2(config)#
S2(config)#copy running-config startup-config
          ^
% Invalid input detected at '^' marker.

S2(config)#exit
S2#
%SYS-5-CONFIG_I: Configured from console by console

S2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

What this step demonstrates: Apply the unused-port shutdown and save the configuration on the second switch.

Tip: **`copy running-config startup-config`** must be run from privileged EXEC mode (**S2#**), not from configuration mode.

Step 15 — Verify port security on S2

Run the same verification commands on S2.

```
show port-security
show port-security address
show port-security interface fastEthernet 0/1
show mac address-table
```

```
S2#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.96da.920e    STATIC      Fa0/2
   1    0001.c935.0702    DYNAMIC     Gig0/1
   1    0090.2b84.3de4    STATIC      Fa0/1
   1    00d0.ffe6.7c3b    STATIC      Fa0/3
S2#show port-security interface FastEthernet 0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0090.2B84.3DE4:1
Security Violation Count   : 0

S2#show port-security address
               Secure Mac Address Table
------------------------------------------------------------------------------
Vlan    Mac Address       Type                          Ports   Remaining Age
                                                                    (mins)
----    -----------       ----                          -----   -------------
   1    0090.2B84.3DE4    SecureSticky                  Fa0/1        -
   1    0001.96DA.920E    SecureSticky                  Fa0/2        -
   1    00D0.FFE6.7C3B    SecureSticky                  Fa0/3        -
------------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 1024
S2#
```

Observations: Fa0/1 shows port security **Enabled**, **Secure-up**, violation mode **Shutdown**, maximum **1** and one sticky MAC (0090.2B84.3DE4). The secure address table contains one **SecureSticky** MAC on each of Fa0/1, Fa0/2 and Fa0/3, and the violation count is **0**.

> **What this step demonstrates:** Confirm the port-security settings and learned sticky MAC addresses on S2.

Part 5 — Test the Network

SSH access and connectivity between the two networks.

Step 16 — Test SSH access (Task 12)

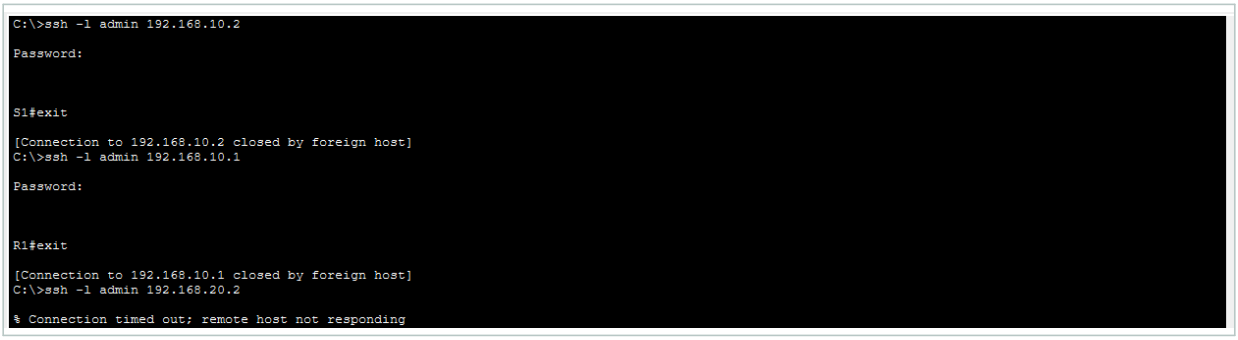
From the PC **Desktop** → **Command Prompt**, connect to the devices using SSH with the **admin** account (the password is typed at the prompt and is not displayed):

From PC1 (PC0 in Packet Tracer)

```
ssh -l admin 192.168.10.2
ssh -l admin 192.168.10.1
```

From PC4

```
ssh -l admin 192.168.20.2
```



The SSH sessions to **S1 (192.168.10.2)** and **R1 (192.168.10.1)** succeed and open directly at the privileged prompt (**S1#**, **R1#**) because the admin user has privilege 15. Typing **`exit`** closes each session.

The attempt to **192.168.20.2** from PC0 returned **Connection timed out; remote host not responding**; the project procedure tests S2 from PC4 on the same network.

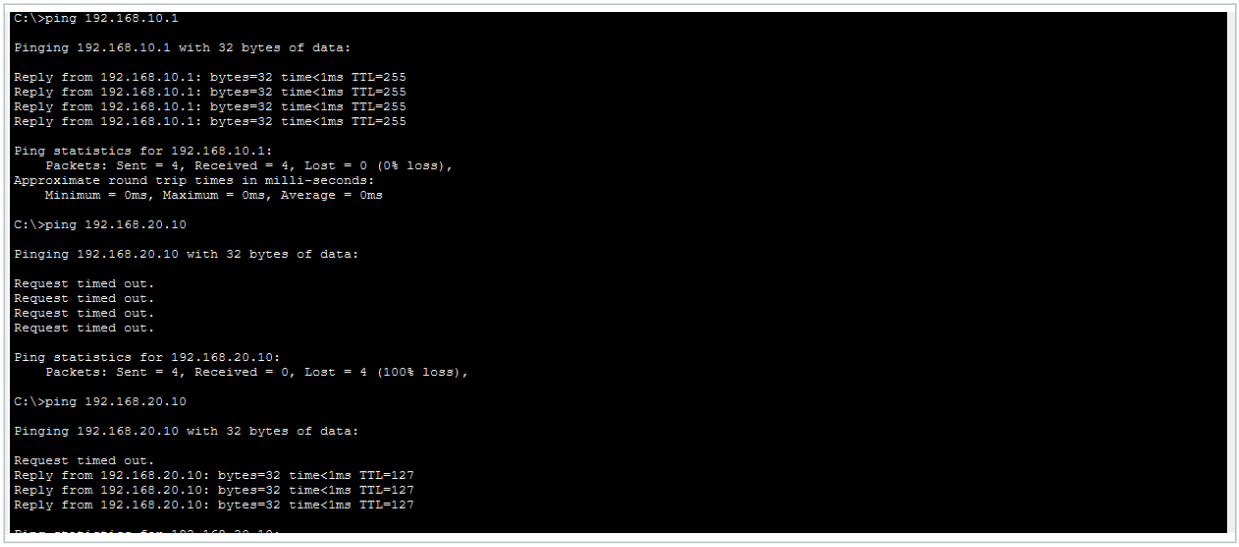
> **What this step demonstrates:** Show that remote administration works over SSH on the router and the switch.

Step 17 — Test connectivity (Task 11)

From the PC command prompt, ping the router gateway and the PCs on the other network:

From PC1 Command Prompt

```
ping 192.168.10.1
ping 192.168.20.10
ping 192.168.20.11
ping 192.168.20.12
```



The ping to the gateway **192.168.10.1** succeeds with 0% loss. The first ping to **192.168.20.10** timed out (100% loss) while ARP and MAC learning completed; repeating it returned replies with **TTL=127**, showing the traffic crosses the router between the two networks.

> **What this step demonstrates:** Prove inter-network connectivity. Expected result: successful replies after ARP/MAC learning has completed.

> **Tip:** The first pings across a router often fail while ARP entries are built. If you see timeouts, repeat the ping.

Part 6 — Unauthorized Device Test

Show that the switch blocks devices it does not trust.

Step 18 — Connect an unauthorized device (Tasks 14 and 15)

To demonstrate port security, first generate traffic from a PC so the switch learns its MAC address (for example, ping the router). Disconnect the PC and connect another device to the same port. Because only one secure MAC address is allowed, the switch detects a violation and shuts the port down. A **Laptop0** was connected to Switch0 for this test.

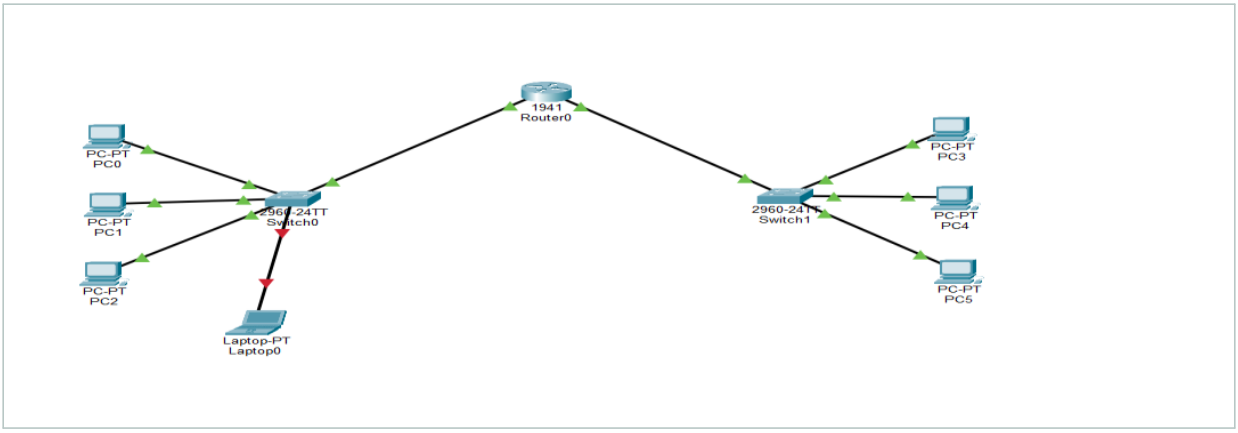
In the topology, both ends of the Laptop0 link turn **red**, which means the switch is not forwarding traffic on that port.

Check the port

```
show port-security interface fastEthernet 0/1
```

Recover the interface

```
configure terminal
interface fastEthernet 0/1
shutdown
no shutdown
exit
```



What this step demonstrates: Demonstrate that an unauthorized device cannot use a secured or unused switch port.

Tip: After a port-security violation the port stays shut down until it is recovered with **`shutdown`** followed by **`no shutdown`**.

Final Verification Checklist

Results observed in the screenshots above.

Test	Expected Result	Observed Result
PC1 → Router ping	Successful	Successful — 0% loss (Step 17)
PC1 → PC4 ping	Successful	Successful after ARP learning, TTL=127 (Step 17)
SSH → R1	Successful	Successful (Step 16)
SSH → S1	Successful	Successful (Step 16)
SSH → S2	Successful	Test from PC4 (the PC0 attempt timed out)
Port security	Enabled	Enabled on S1 and S2 (Steps 13, 15)
Sticky MAC learning	Successful	One SecureSticky MAC per port Fa0/1–3
Unauthorized MAC test	Violation detected	Laptop0 link shows red (Step 18)
Unused ports	Shutdown	Fa0/4–24 administratively down (Steps 12, 14)

Conclusion

This mini project gave practical experience in securing Cisco routers and switches. Basic IP connectivity was configured between two networks, console and privileged EXEC access were protected, SSH was enabled for secure remote administration, locally configured passwords were encrypted, and switch port security was used to restrict unauthorized devices. Together these controls improve the security of a small campus network.